R-connect API – v2.0

Author: Robopac R&D Dept. | Revision: v.2 | Date: 10/03/2026

Rconnect WS3 API Technical Specifications

Notice:

*Robopac does not guarantee uninterrupted availability of the API services. The services may be temporarily unavailable due to maintenance, updates, or technical issues.

**Robopac may introduce changes to the API, including new features, modifications, or deprecations. While efforts are made to maintain backward compatibility, clients are responsible for adapting their integrations to newer API versions.

***The contents of this document are subject to revision without notice due to continued progress in methodology, design and manufacturing. Robopac shall have no liability for any error or damage of any kind resulting from the use of this document.

Sommario

1. Introduction 3
2. Authentication 4
2.1 Access Token Request 4
2.2 Refresh Token Request 5
3. Version Check and Machine Schema 6
3.1 Service Status Check 6
3.2 Machine Schema Request 6
4. Realtime Data 7
4.1 Realtime Data Request 7
4.1.1 Example with Multiple Stream Filter 7
4.1.2 Example with Specific Signal Filter 8
4.1.3 Combined Example 9
5. Historical Data 9
5.1 Historical Data Request with Initial Date 9
5.2 Historical Data Resume 10

1. Introduction

This document describes **version 2.0 of the R-connect platform APIs**.
Each section includes a description, endpoint, parameters, and request/response examples.

2. Authentication

2.1 Access Token Request

```
POST https://rconnectws3.robopac.com/oauth/token
```

Requests or refreshes an access token.
The endpoint supports two grant_type values:

- password
- refresh_token

EXAMPLE REQUEST:

```
POST https://rconnectws3.robopac.com/oauth/token


{
  "grant_type": "password",
  "username": "user@example.com",
  "password": "mypassword",
  "client_id": "019960ef-71be-73fd-a368-49a1264f8025",
  "client_secret": "mysecret",
  "scope": "get_realtime get_history"
}
```

EXAMPLE RESPONSE:

```
200 OK
{
  "token_type": "Bearer",
  "expires_in": 7200,
  "access_token": "eyJ0eXAiOiJKV1QiLCJhbGciOi...",
  "refresh_token": "def5020053408af29e5af74d5447ccde..."
}
```

2.2 Refresh Token Request

```
POST https://rconnectws3.robopac.com/oauth/token
```

Request with grant_type=refresh_token. Username and password are not required.

EXAMPLE REQUEST:

```
POST https://rconnectws3.robopac.com/oauth/token
{
  "grant_type": "refresh_token",
  "refresh_token": "{{refresh_token}}",
  "client_id": "{{client_uuid}}",
  "client_secret": "{{client_secret}}",
  "scope": "get_realtime get_history"
}
```

EXAMPLE RESPONSE:

```
200 OK
{
  "token_type": "Bearer",
  "expires_in": 7200,
  "access_token": "eyJ0eXAiOiJKV1QiLCJhbGciOi...",
  "refresh_token": "def502006b6aa74c21fee5e0b7993e189064993c..."
}
```

If API calls are made without authentication (except for the version and service status check), the web service returns:

```
401 Unauthorized

{

    "success": false,

    "error": "Authentication Error",

    "message": "Token is missing, invalid or expired"

}
```

3. Version Check and Machine Schema

3.1 Service Status Check

```
GET https://rconnectws3.robopac.com/api/isAlive/api/isAlive
```

Checks API availability and returns version and status information.

EXAMPLE RESPONSE:

```
200 OK

{

    "app_version": "12.1",

    "app_datetime": "2025-09-24 09:38:20",

    "datetime": "2026-03-10 08:08:25",

    "mode": "mode1"

}
```

3.2 Machine Schema Request

```
GET https://rconnectws3.robopac.com/api/schema/{chiave_impianto}
```

Returns the schema of available streams and field references.

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
    "anagrafica": { ... },
    "streams": { "1": ["S01C00","S01D01"], "11":
["S11C00","S11D01"] }
```

```
  }
}
```

4. Realtime Data

4.1 Realtime Data Request

```
{url} = https://rconnectws3.robopac.com
```

```
GET {url}/api/realtime/{chiave_impianto}
```

Returns the machine **realtime data**. Available filters:
- sninv[]: filters by stream
- fieldref[]: filters specific signals

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
    "anagrafica": { ... },
    "streams": {
      "1": { "data": { "COMMAND WORD": {"value": "1"} } },
      "11": { "data": { "Cycle Type": {"value": "0"} } }
    }
  }
}
```

4.1.1 Example with Multiple Stream Filter

```
GET {url}/api/realtime/{chiave_impianto}?sninv[]=1&sninv[]=11
```

EXAMPLE RESPONSE:

```
200 OK

{
  "impianto": {
    "streams": {
      "1": { "data": { "R-connect Power": {"value": null},
"COMMAND WORD": {"value": "1"}, "time_stamp": "2025-09-19
06:30:16" } },
      "11": { "data": { "Recipe in use": {"value": "3"}, "Speed":
{"value": "80"}, "time_stamp": "2025-09-19 08:06:41" } }
    }
  }
}
```

4.1.2 Example with Specific Signal Filter

```
GET
{url}/api/realtime/{chiave_impianto}?fieldref[]=S01C00&fieldref[]
=S11D12
```

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
    "streams": {
      "1": { "data": { "Z-UMTS Version": {"value": "600"},
"time_stamp": "2025-09-19 06:30:16" } },
      "11": { "data": { "Cycle Type": {"value": "0"},
"time_stamp": "2025-09-19 08:12:13" } }
    }
  }
}
```

4.1.3 Combined Example

Stream **9** with specific fieldref filters. Data are grouped by stream.

```
GET
{url}/api/realtime/{chiave_impianto}?sninv[]=9&fieldref[]=S01C00&
fieldref[]=S11D12
```

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
    "streams": {
      "9": { "data": { ... } },
      "1": { "data": { "Z-UMTS Version": {"value": "600"} } },
      "11": { "data": { "Cycle Type": {"value": "0"} } }
    }
  }
}
```

5. Historical Data

5.1 Historical Data Request with Initial Date

Returns batches of **up to 60 elements** and a next_poll pointer.

```
GET {url}/api/history/{chiave_impianto}?datefirst=2025-09-17
00:00:00&sninv[]=9&fieldref[]=S01C00&fieldref[]=S11D12
```

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
```

```
    "streams": {
      "1": {
        "data": [
          { "Z-UMTS Version": {"value": "600"}, "time_stamp":
"2025-09-17 06:03:42" },
          { "Z-UMTS Version": {"value": "600"}, "time_stamp":
"2025-09-17 06:20:43" }
        ],
        "next_poll": "4MCjR1mw"
      },
      "11": {
        "data": [
          { "Cycle Type": {"value": "0"}, "time_stamp": "2025-09-
17 05:19:33" }
        ],
       "next_poll": "lrC9oEoD"
      }
    }
  }
}
```

5.2 Historical Data Resume

Historical data retrieval can continue using the next_poll tokens until exhaustion.

```
GET
{url}/api/history/{chiave_impianto}?fieldref[]=S01C00:4MCjR1mw&fi
eldref[]=S11D12:lrC9oEoD
```

EXAMPLE RESPONSE:

```
200 OK
{
  "impianto": {
    "streams": {
      "1": {
        "data": [ { "Z-UMTS Version": {"value": "600"},
"time_stamp": "2025-09-19 08:21:35" } ]
      },
      "11": {
        "data": [ { "Cycle Type": {"value": "0"}, "time_stamp":
"2025-09-17 12:51:18" } ]
```

```
            }
        }
    }
}
```